Compositional Neural Mechanics with Observable Physical Uncertainty

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Abstract

A reconstruction can reproduce observed motion while predicting the wrong response to a new interaction. We study a representation that separates reusable bulk material energies from relational attachment energies, and composes both through a nonlinear equilibrium solver. Scene-specific continuous descriptors encode material heterogeneity, support stiffness, nonlinearity, location and extent. Prescribed tool positions and orientations are inputs; deformation and reaction forces are outputs. We distinguish forward-law transfer with known descriptors from inference with unknown descriptors, and distinguish numerical convergence from uncertainty calibration.

The attachment-only model preserves an exact common-stiffness scaling symmetry: with zero body load, scaling all stiffnesses changes forces but leaves every position-controlled equilibrium unchanged. Known gravity and density can supply the missing load reference. Development experiments demonstrate accurate nonlinear response, improved inverse optimization, and useful refinement from an additional observation. On six withheld-composition objects and six draws on new geometries, shared-law position error is 0.0181 and 0.0183 mm, versus 3.09 and 3.29 mm for the stabilized 30,000-step custom direct baseline with the same oracle descriptors. On six separate objects with 15 hidden continuous descriptors, one additional stretch observation reduces shared-law motion error from 0.339 to 0.080 mm and force error from 0.096 to 0.029 N without force measurements. The adaptive policy ties fixed stretch; nominal uncertainty remains sensitive to model discrepancy. Frozen laws also compose with normal contact in two known-descriptor development cases. A separate three-parameter continuous numerical reference passes declared surrogate-validation and integration checks. The present scope is quasistatic tracked-point mechanics with known geometry and categorical law families; it does not establish a complete video-to-simulator system or superiority to published vision systems.

1. Scientific question

The target is counterfactual physical accuracy: given an observation history, can a model predict the deformation and reactions under a previously unseen command? Rendering agreement measures a different quantity. Multiple physical explanations can produce the same visible shape, and a better image fit need not reduce error in future forces. We therefore report physical outputs separately and evaluate observations through their effect on future predictions.

Contribution under test

We test whether shared local laws, composed by the same mechanical operator, transfer across material-support combinations and geometry more systematically than direct displacement prediction. A second question is whether scene inference and action-conditioned refinement preserve physical ambiguity while improving useful predictions. Neural constitutive laws, differentiable mechanics, Gaussian support fields and active sensing are established ingredients; the proposed contribution is their composition and intervention-based validation.

2. Representation and mechanics

Material coordinates X remain attached to the object. Its state is x(X)=X+u(X), with deformation gradient F=I+grad u. Geometry is a mapped cuboid with optional taper and arch. The present material field is isotropic and hyperelastic; spatial variation is expressed in material coordinates rather than a fixed world-space field.

```
W(F, [s,z]) = exp(s) W_shared(F,z)
P = dW/dF
```

The bulk network uses rotational invariants and positive coefficients. Factoring out the common stiffness exp(s) enforces the scaling identity exactly. This construction provides objectivity and a consistent energy-stress relation; we do not claim a general polyconvexity guarantee. The training families are Mooney-Rivlin and Yeoh-like responses, with continuous scale and shape descriptors.

A separate shared attachment energy depends on relative displacement between a material point and its commanded tool target. Gaussian footprints specify location and width. The reference-area-normalized energy includes positive stiffness and nonlinear shape parameters, and supports polynomial and saturating response families. These are bilateral compliant attachments, not nonpenetrating contact constraints.

```
Pi(u) = integral W(F,z_m) dV
+ sum integral psi(delta(u,a),z_b) dA
- integral rho g . u dV
R(u,z,a) = dPi/du = 0
```

A quadratic finite-element discretization, Newton iterations and line search solve equilibrium. The implementation checks residuals and positive deformation Jacobians. Gravity and density remain explicit body-load inputs. Forces and torques on the prescribed tools are derived from the same energy and solved state.

Scene inference

The 15 continuous unknowns are bulk log stiffness, shape and heterogeneity amplitude; two attachment log stiffnesses and nonlinearities; four footprint center coordinates; and four log widths. Geometry, categorical bulk/interface family and heterogeneity pattern are conditioned on. The learned law weights stay fixed while these scene descriptors are fitted to noisy tracked surface positions. Reaction observations enter only in explicitly labeled measurement variants.

```
H du/dz = -dR/dz
L(z) = 0.5 ||r_observation(z)||^2 + 0.5 ||r_prior(z)||^2
```

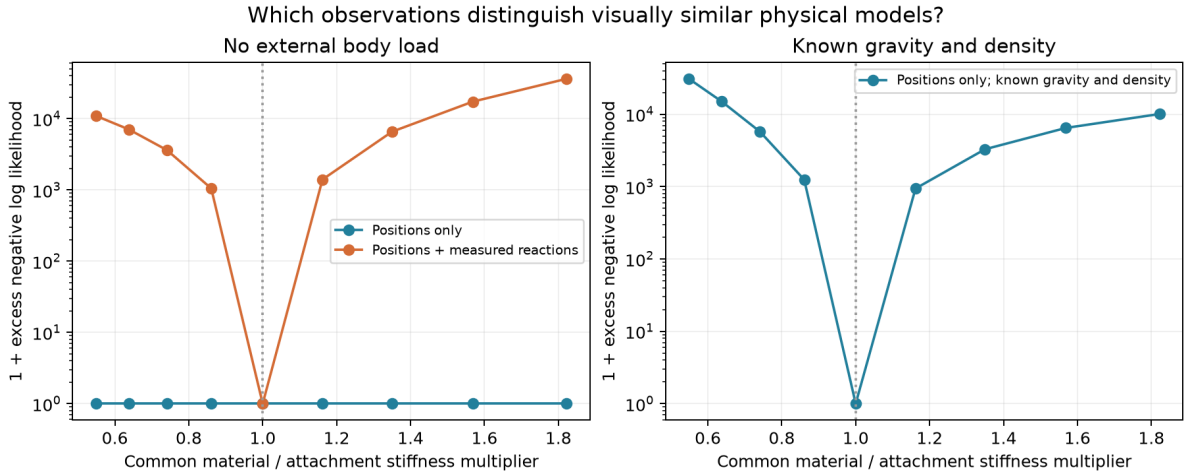
Implicit sensitivities differentiate the converged mechanical state. A bounded Gauss-Newton trust-region solver uses the full residual Jacobian and its scaling to improve optimization conditioning. Its stopping status is reported separately from parameter accuracy, predictive accuracy and posterior validity.

3. Observability and refinement

Common-scale ambiguity. In the attachment-only model with no additional fixed-stiffness energy terms, if gravity and other prescribed body forces vanish, multiplying every bulk and attachment stiffness by c multiplies the entire energy by c. The equilibrium displacement is unchanged, whereas the reactions multiply by c. All rendered views of that state are therefore unchanged. This ambiguity persists across every prescribed-position experiment in this model; additional cameras cannot remove it.

With fixed-penalty contact, this identity requires co-scaling the penalty or imposing ideal hard-contact constraints. Numerical penalty stiffness is not physical scale evidence. The contact study uses known gravity and density and does not infer contact.

Known gravity with known density introduces a fixed reference load and changes this conclusion: stiffness affects visible sag. If density is also free to scale with stiffness, a joint ambiguity remains. These assumptions matter when claiming that position-only observations identify physical properties.



Development likelihood slices at fixed relative stiffnesses. These are not marginalized posteriors or confidence intervals. The zero-gravity and known-gravity experiments use different noise seeds.

Functional uncertainty

A local Gaussian approximation uses the observation Jacobian and an explicit prior precision. Candidate actions are ranked by their expected reduction in variance of a specified future response, rather than total parameter variance. The target commands are public inputs; their response labels are unavailable during selection.

```
C = (J_obs^T J_obs + P_prior)^(-1)
C_after(a) = (C^(-1) + J_a^T J_a)^(-1)
score(a) = trace[A (C - C_after(a)) A^T]
```

This approximation does not automatically account for nonlinear multimodality, active bounds or model discrepancy. The development bands set model discrepancy to zero and are labeled nominal. A separate small reference posterior tests integration convergence before any claim that its intervals are trustworthy. Empirical coverage and fixed/random acquisition controls remain required.

4. Development evidence

Forward model	Position RMSE (mm)	Force RMSE (N)	Torque RMSE (Nm)
Shared laws	0.0190	0.00704	0.000099
Neo-Hookean	0.0494	0.04496	0.000252

Four validation objects, six commands each; both models use known descriptors and n=4. One original runtime failure was replayed in an isolated process with identical settings; the failure record is retained. These validation numbers are not the reserved test.

Object / model	Original position (mm)	Gauss-Newton (mm)	GN force (N)
0 / analytic	0.0092	0.0088	0.00245
0 / learned	0.0101	0.0091	0.00345
1 / analytic	0.0243	0.0190	0.00256
1 / learned	0.0250	0.0219	0.00225

Full 15-parameter development inversion, two observed objects and two future commands each. All four Gauss-Newton fits stopped successfully by ftol, versus zero original fits within their budgets. All motion errors decreased; force errors increased for object 0 and decreased for object 1. This is an optimization comparison, not an independent generalization result.

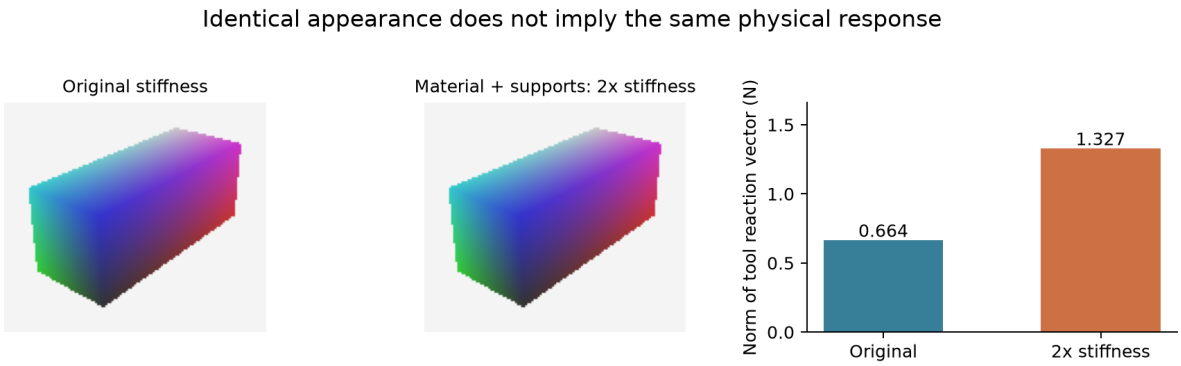
Observed failure modes and repairs

A narrow-slip attachment training distribution produced a 4.065 mm same-mesh error in an exposed stretching case. A separately documented warm-start, wide-slip training run reduced that case to 0.079 mm. A failed alternative wide-range run is retained. The repaired law was frozen before reserved testing; the exposed case is not counted as evidence of unseen-scene transfer.

Development mesh checks also showed that n=4 was inadequate for several extreme reference cases; targeted n=6 to n=8 checks passed with field differences of 2.98-8.15 micrometres. The reserved benchmark consequently uses a fixed n=8 analytic reference. That choice is supported by development tests, not individual convergence certification for every reserved scene.

4.1. Visual agreement and physical response

A deterministic calibrated renderer uses the same solved mechanical field as the force and position predictions. Surface color is fixed in material coordinates, with no learned image correction. We evaluate three fixed views using foreground-union RGB error, joint-visible depth error and silhouette intersection-over-union (IoU); missing and extra pixels remain recorded separately.



Zero gravity; prescribed positions; same equilibrium. All three full RGB/depth views are identical.

Exact zero-body-load counterexample: co-scaling bulk and support stiffness preserves equilibrium and all three full RGB/depth views while doubling reactions. Display crops are identical. This is a controlled mathematical ambiguity, not a claim about every gravity-loaded scene.

Object / prediction	Position mm	Force N	RGB RMSE	Depth mm	IoU
0 / initial	0.2017	0.02958	0.0424	0.5927	0.9940
0 / after stretch	0.1135	0.00568	0.0142	0.1673	0.9989
1 / initial	0.7694	0.06666	0.0562	0.9377	0.9849
1 / after stretch	0.0813	0.01510	0.0234	0.1269	0.9980

Two development objects, one declared held command per object. Analytic n=4 references versus learned n=2 predictions include discretization/model error, so these values differ from the earlier same-n=2 pilot. View means are descriptive; pixels are correlated. RGB is normalized to [0,1].

Both objects improve in rendered and physical outputs after an extra stretch observation. Even the initial silhouettes have high IoU, illustrating why a favorable appearance metric cannot substitute for force or deformation validation. This bridge uses known geometry and synthetic appearance; it does not evaluate visual reconstruction, camera estimation or inference directly from images.

5. Reserved forward transfer

Training uses 54 objects from 18 combinations of geometry, material family, support layout and heterogeneity, with six commands per object, plus four validation objects. Six factor combinations are withheld while each pair of factors remains represented. The interface law family is sampled independently of these four factors. The reserved test has six withheld-composition objects and six draws on two new geometries, each evaluated under four new command directions or combinations. Test identities, checkpoints, source hashes and methods are committed before generating results.

All forward methods receive oracle descriptors. Comparisons include the matched analytic family at n=4, small-strain-matched Neo-Hookean, learned laws at n=4, and custom direct point predictors trained for 6,000 and 30,000 steps. The direct predictors receive numeric law fingerprints from the same learned laws; this provides a shared physical prior but is not computationally identical to full solver access. These custom networks are not reproductions of PhysTwin or MeshGraphNet.

Runtime and preprocessing provenance

The original forward run completed 240 of 288 reference/prediction rows; eight scene workers reached their 1,800-second limits. A JSON tuple/list comparison then blocked aggregation. Calling the unchanged aggregator with committed JSON records preserved all 288 original rows, including 48 missing-output failures. A separately audited recovery recomputed only those 48 missing calls in a new directory, using identical frozen sources, checkpoints, n=4/n=8 discretizations, commands and the original runtime. All 288 merged rows are now complete; the original failure ledger remains available.

A source-only normalization review found that command coordinates absent from training were divided by a standard-deviation floor of 1e-8. Separate stable controls replace only exactly zero training inputs with fixed physical-unit scales; weights, target normalization and all 324 training predictions remain unchanged. The two stable checkpoints were frozen before reference scoring. However, all 96 initial forecast writes failed because nested output directories were absent. A directory-only replay later produced 96 finite artifacts with the same frozen prediction code and checkpoints, after primary outcomes were visible. No retraining, preprocessing redesign or model selection followed those outcomes. Successful pre-truth forecast commitment is not claimed.

The fair direct comparison uses these successful stable controls. The original unstable baselines are retained as preprocessing diagnostics, not evidence for an architectural advantage. Stable controls run on CPU with PyTorch 2.4; the original/recovered oracle models run on GPU with PyTorch 2.14. No cross-runtime bit-identity claim is made.

5.1. Audited forward results and stable controls

Cohort / model	Position mm	Nonrigid mm	Force N
composition / Matched analytic	0.0164	0.0156	0.00139
composition / Neo-Hookean	0.0295	0.0222	0.02889
composition / Shared laws	0.0181	0.0168	0.00688
composition / Direct 6k, stable	3.6504	2.2350	0.72036
composition / Direct 30k, stable	3.0888	1.7861	0.38040
geometry / Matched analytic	0.0159	0.0150	0.00134
geometry / Neo-Hookean	0.0298	0.0220	0.02348
geometry / Shared laws	0.0183	0.0166	0.00367
geometry / Direct 6k, stable	3.7504	2.1410	0.58107
geometry / Direct 30k, stable	3.2933	1.9063	0.46042

Each cohort has six objects and four commands per object. All 24 commands per method are included. Position is coordinate RMSE; nonrigid error removes the best rigid alignment. Means weight objects equally. The geometry cohort contains six draws conditional on two new shape instances, not six independent geometry families. The matched analytic n=4 control also differs from the n=8 reference.

Both stable direct budgets are reported without selecting the more favorable comparison. Shared laws retain a substantial advantage over the corrected custom direct predictors in this oracle-descriptor experiment. This is controlled evidence for composing reusable laws through mechanics; it is not a state-of-the-art vision benchmark or an official PhysTwin/MeshGraphNet comparison.

Preserved unstable-normalization diagnostic

Cohort	Original direct 6k mm	Original direct 30k mm
composition	543604.2	136568.2
geometry	544037.4	136593.5

These extreme values diagnose the frozen zero-variance preprocessing defect. They are intentionally excluded from the fair architectural comparison above. All original outputs, the failed-write ledger, and both directory-only and missing-call recovery records remain preserved.

6. Observation refinement and uncertainty checks

Policy	Mean position (mm)	Mean nominal 90% coverage
initial	0.3708	98.4%
fixed_stretch	0.0170	85.3%
fixed_right_bend	0.1119	92.0%
fixed_opposed_twist	0.0295	81.4%
random	0.0235	80.4%
target_task_variance	0.0170	85.3%

Two exposed development objects and two future commands per object. Each non-initial policy receives exactly one additional observation. The variance policy selected stretch for both objects and therefore ties fixed stretch. Pointwise coordinates are correlated; these coverage fractions are descriptive, not independent calibration trials or evidence of a superior policy.

6.1. Unseen objects with unknown continuous parameters

Six fresh continuous scene draws cover both bulk and attachment families on three familiar geometries. All 15 continuous descriptors are hidden; geometry, categorical families, gravity and density are known. Observations come from n=4 analytic simulations, while both fitted models use n=2. Every probe policy receives the initial position packet and exactly one additional packet; forces are not observed.

Policy	Analytic mm	Shared mm	Analytic coverage	Shared coverage
initial	0.4268	0.3395	97.1%	97.0%
fixed_stretch	0.0805	0.0802	34.3%	33.0%
fixed_right_bend	0.2188	0.2163	43.7%	44.7%
fixed_opposed_twist	0.0826	0.0835	48.3%	47.8%
uniform_random	0.0954	0.0973	47.2%	46.6%
target_task_variance	0.0805	0.0802	34.3%	33.0%

All 144 policy-command predictions are retained. 41 of 48 fits met their stopping criteria; 7 reached the evaluation limit. All 12 stretch fits selected across the two models converged. Incomplete fits are not silently excluded from metrics. Coverage is marginal coordinate coverage of nominal 90% local Gaussian bands; coordinates and commands are correlated.

The variance policy selected stretch for every object in both models, so it exactly ties that fixed control. Shared-law error after stretch is approximately 0.080 mm, close to fitted analytic mechanics. Both improve on all six objects relative to their initial estimates. This supports observation-based refinement, but not an adaptive advantage over the strongest fixed action.

Uncorrected bands become sharply overconfident after probing: the learned model covers only about one third of target coordinates. A separately frozen validation-only discrepancy estimate is evaluated as an additive prediction variance, with coverage, width and interval score reported together. It changes neither fitted parameters nor actions and is not a posterior-calibration guarantee.

Model	Coverage %	Half-width mm	Interval score mm
Analytic	34.3 / 95.7	0.026 / 0.188	0.729 / 0.432
Shared	33.0 / 96.2	0.026 / 0.196	0.741 / 0.442

Before / after adding the prospectively frozen validation MSE. Lower interval score is better; it penalizes both excessive width and missed targets. These six-object descriptive results do not establish population calibration.

6.2. Reaction prediction from position observations

A prospectively specified secondary replay evaluates reactions for the same fitted parameters and policy choices. No fitting or selection is repeated and no force observations are added. Reference and prediction torques use the same declared tool origins. All 108 replay states and 144 policy-command reaction rows completed and passed independent checks.

Policy	Analytic force N	Shared force N	Analytic torque Nm	Shared torque Nm
initial	0.11247	0.09605	0.003035	0.002384
fixed_stretch	0.02839	0.02921	0.000559	0.000525
fixed_right_bend	0.04660	0.04278	0.001215	0.001186
fixed_opposed_twist	0.02799	0.02851	0.000420	0.000421
uniform_random	0.03319	0.03429	0.000507	0.000508
target_task_variance	0.02839	0.02921	0.000559	0.000525

Mean coordinate RMSE, equally averaged over two held commands per object and six objects. Primary fit-limit failures remain in the ledger; successful reaction replay does not make an incomplete parameter fit converged.

The learned model reduces reaction-force error from about 0.096 N initially to 0.029 N after stretch. Fixed twist has slightly lower force and torque errors than the selected stretch on this cohort. The acquisition criterion targets future positions; it is not a universally optimal criterion for every physical output. Fitted analytic and shared-law models have similar post-probe performance.

6.3. Numerical uncertainty and contact qualification

Reference posterior

The completed extension evaluated 2,801 unique mechanics nodes with 0 failed nodes. Its independent audit passed 18,566 checks of saved responses, provenance and reported convergence decisions. The position-only V reference passes the declared numerical consistency tests; the position-plus-wrench VF reference does not.

For VF, doubling the proposal width changes an interval endpoint by 0.146499 posterior standard deviations, above the fixed 0.05 limit, despite stable moments and evidence. The conditional scale CDF is accurately integrated, but the remaining two-dimensional ratio quadrature resolves sharp tail transitions poorly. No converged VF interval or posterior-calibration claim is made. Earlier failures remain preserved.

This reference calculation exploits the exact zero-gravity common-scale identity: it solves in two relative stiffness coordinates and integrates the common scale separately with the original bounded prior. Grid refinement, conditional integration and proposal-width checks are required. It is a diagnostic of a small conditional inverse problem, not a 15-dimensional posterior validation.

Separate continuous numerical reference

A separately planned repair fits a degree-four polynomial to cached forward-response vectors, with a degree-three crosscheck. It retains the original prior, likelihood and exact common-scale wrench law. All 64 fresh exact FEM validations pass, including samples from both proposal widths and an outer shell. Continuous joint scrambled-Sobol importance integration uses actual proposal-density correction and weighted empirical quantiles, rather than Gaussian intervals.

Across fixed sample levels, proposal widths, scrambles and polynomial degrees, the largest endpoint change is 0.003525 posterior SD, below 0.05. Independent review recomputed all fresh response errors and the primary 1,048,576-sample integration. This qualifies the declared developmental numerical surrogate checks; it does not retroactively qualify the original Gauss-Hermite intervals.

Log parameter	Mean	90% numerical interval
Bulk scale s	0.251191	[0.245468, 0.256903]
Left support log K	5.861011	[5.839067, 5.883025]
Right support log K	6.675390	[6.660346, 6.690457]

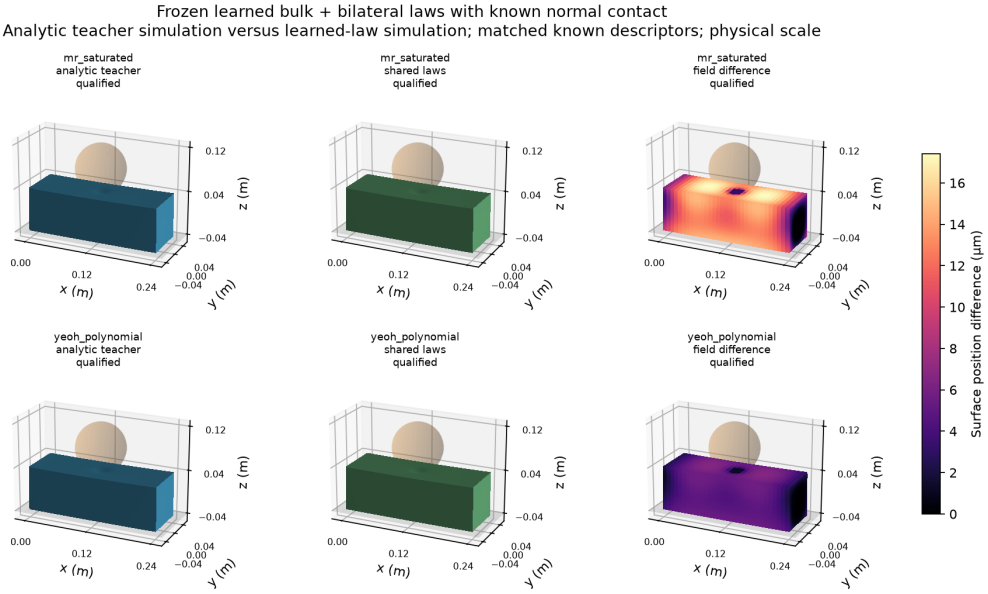
Tail checks are empirical: the primary proposal samples no radius-above-eight points, while wider proposals supply finite tail samples. Neither these estimates nor shell validation are a global error certificate. This is a three-parameter problem with known nuisance values, not exact-FEM posterior certification, population coverage or validation of the full 15-dimensional inference.

6.4. Frozen-law transfer to normal contact

A separate development experiment composes the frozen learned bulk and nonlinear bilateral attachment laws with a known analytic normal-penalty operator for a floor and sphere, under gravity. No network retraining is performed. Both material/interface combinations use known descriptors. All 16 analytic/shared equilibrium states completed; all four model-family mesh and contact-quadrature qualification sets passed, independently checked in 265 audit checks.

Bulk + attachment	Field RMS (um)	Sphere force error (N)	Sphere force error (%)
Mooney-Rivlin + saturating	6.281	0.015485	2.019
Yeoh + polynomial	2.397	0.006356	0.841

Shared-law versus analytic-teacher differences on the finest n8/q32 states. These are descriptive transfer errors, not thresholds used to qualify the reference. Fixed-penalty mesh and quadrature consistency do not prove continuum convergence. Earlier coarse and tiny-contact failures remain preserved.



Known geometry, oracle descriptors and static equilibrium only. Contact is normal and frictionless; rigid obstacle poses are prescribed. Finite penalty penetration remains. This does not establish unknown-contact inference, dynamic impact, soft-soft interaction or public-dataset improvement.

7. Relation to vision-based physical reconstruction

PhysTwin [1] reconstructs deformable digital twins from sparse multiview videos using shape priors, spring-mass mechanics and Gaussian appearance. Our current tracked-point, known-geometry study isolates local-law composition and inference; it is not an end-to-end comparison with that reconstruction system.

PhysGS [2] maintains Bayesian physical-property beliefs over Gaussian representations and refines them from observations. Our evaluation instead targets future mechanical responses and explicit observability under a coupled equilibrium model. Bayesian property estimation itself is not a novel ingredient here.

PhysHO [3] incorporates neural constitutive modeling into human-object interaction simulation, while **NeuSpring [4]** uses neural spring-property fields. These works preclude claiming that a neural material field or constitutive component alone is new. Our hypothesis concerns reusable bulk/interface composition and its validation under unknown scene descriptors and new interventions.

8. Scope and next evidence

The implementation supports useful pieces of the proposed architecture: reusable bulk and attachment laws, position-controlled equilibrium, scene inference, structural observability and task-aware action scoring. Current evidence is restricted to mapped cuboids, known categorical families, known geometry and quasistatic elasticity. It does not establish arbitrary unseen material families, topology changes, soft-soft contact, friction, impacts, learned perception, Gaussian rendering quality or new-view acquisition.

The six-object inverse study now provides evidence of observation-based refinement under hidden continuous material and support descriptors. Its fixed-action ties, incomplete fits and overconfident uncorrected bands remain part of the result. The completed oracle forward study includes both stabilized direct budgets and explicitly preserved runtime/preprocessing failures. Secondary reaction replay confirms that position-only refinement also improves force predictions, while fixed twist is slightly better for reactions than the position-targeted selected stretch. Two known-descriptor normal-contact compositions pass fixed-penalty numerical checks without retraining. The original VF reference remains unresolved at its ratio quadrature; a separately validated continuous numerical surrogate passes finite consistency checks with empirical, not globally certified, tail evidence. Existing repository benchmark results remain preserved but cannot be attributed to this new representation without adapting and rerunning it.

References

[1] H. Jiang et al. PhysTwin: Physics-Informed Reconstruction and Simulation of Deformable Objects from Videos. ICCV 2025. Primary source.

[2] S. Chopra et al. PhysGS: Bayesian-Inferred Gaussian Splatting for Physical Property Estimation. CVPR 2026. Primary source.

[3] S. Jiang and G. H. Lee. PhysHO. CVPR 2026. Project and paper. Primary source.

[4] Q. Xu et al. NeuSpring. arXiv:2511.08310, 2025. Primary source.